higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

PROCESS TECHNOLOGY
NQF LEVEL 3

26 FEBRUARY 2019

This marking guideline consists of 5 pages.

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QUESTION 1

1.1 False
1.2 False
1.3 True
1.4 False
1.5 True

(5 × 1) **[5]**

QUESTION 2

2.1 H
2.2 J
2.3 A
2.4 I
2.5 G

(5 × 1) **[5]**

QUESTION 3

3.1 3.1.1 Property of a fluid to resist flow (2)

3.1.2 Detailed document that helps to select chemical products,✓ handle chemicals safely✓ and act correctly in an emergency✓ (3)

3.1.3 Process of selecting a small representative portion of material from a batch or stream to see what is happening in the system (2)

3.1.4 Measure of mass per unit volume (2)

3.2
- Flush the sample container to remove all traces of other gases and air present.
- The sampling technique should not change the properties of the gas.
- Sample containers should not affect the physical and chemical properties of the sample.
- Some gases may create safety hazards if sufficient quantities are released to the atmosphere, e.g. fire, explosives, etc.
- Do not expose samples to flames, fire, sparks, heat or static electricity that may cause gases to ignite.
- Sampled material may need to be kept at a specific temperature or pressure to preserve its condition by using correct procedures. (6)

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3.3 $n = m/M$✓
$= 16\ g/100\ g.mol^{-1}$
$= 0,16\ mol$✓

$C = n_{CaCO3}/V$✓
$= 0,16\ mol/75\ dm^3$✓
$= 2,1333\ mol/dm^3$✓ (5)
[20]

QUESTION 4

4.1 A – Shell-side fluid in
B – Baffle
C – Shell
D – Tube sheet
E – Outlet plenum
F – Out
G – In
H – Tube-side fluid
I – Inlet plenum
J – Shell-side fluid out
K – Tube bundles with U-tube
L – Shell side (12)

4.2 It does not require any cooling water from a cooling tower. (2)

4.3 4.3.1 It increases heat transfer from the hot fluid in the tubes to the air flow surrounding the tubes.

4.3.2 It controls the rotational speed of the fan blades.
(2 × 2) (4)

4.4 Cool water is pumped from the basin at the bottom of the cooling tower to the heat exchanger.✓ The heat exchanger cools the hot process fluid to the cooling water.✓ The process fluid then goes for further processing or storage✓ while the cooling water, which has picked up heat from the process fluid, flows to the top of the cooling tower.✓ From the top of the tower, the hot water falls down through the tower exposing it to air✓ causing the water to cool.✓ The cooled water collects at the bottom of the tower from where it is pumped to the heat exchanger for reuse.✓ Heat transfer taking place in a cooling tower happens through evaporation.✓ The heat escapes with the vapour causing the remaining water to cool.✓ (9)

4.5
- Oxygen (21%)
- Nitrogen (78%)
- Other gases (1%) (3)

4.6 CH_4✓ + $2O_2$ (g)✓ ⟶ CO_2 (g)✓ + $2H_2O$ (g)✓ (4)
[34]

QUESTION 5

5.1 5.1.1 Process of removing liquid from products in solid, liquid or gas phase

5.1.2 Solid materials that take changed ions from water and release an equal number of ions in water

5.1.3 Process where molecules stick on the surface

5.1.4 Process that removes a solute from a solution through a semipermeable membrane

(4 × 2) (8)

5.2 It controls the humidity levels by allowing the user to program a desired humidity value. (2)

5.3 The solution to be dried is fed into an atomiser which is at the top or bottom of the chamber.✓ The droplets and hot gas are sprayed into the chamber✓ in the same direction✓ to allow the gas to interact with the liquid while it is still hot.✓ This allows the droplets to be evaporated efficiently as possible✓ and also allows powder to form.✓ The powder is then removed from the chamber through a pipe.✓ The slightly cooled gas with the powder that did not drop through is removed through an outlet connected to separation equipment.✓ (8)

5.4 5.4.1 It occurs when a material sticks to itself forming clumps in the fluid which can result in blockages or decrease the efficiency of the drying process. (2)

5.4.2 Blockages and malfunctioning of dryers (2)

5.4.3
- Inadequate drying of the product due to temperature
- Incorrect concentration of wet material or mixture
- Incorrect material flow rate through the dryer (Any 2 × 1) (2)

5.4.4
- Ensure that the correct temperature is applied to the dryer.
- Ensure that the wet material is mixed correctly with the correct concentration.
- Ensure that the material moves through the dryer at the correct flow rate. (3)

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5.5 The flocculated raw water is at the bottom of the flotation tank.✓ Air is dissolved under pressure in a small amount of the water in a saturator.✓ The water flow is called the saturation or recirculation flow.✓ The dissolved air comes out of this saturated solution in the form of bubbles✓ because of the pressure drop when the saturated flow is added to the influent or flocculated raw water to be treated.✓ The rising velocity of these air bubbles is greater than the water velocity resulting in the collision of the air bubbles with flocs.✓ The attachment or entrapment of the flocs within the air bubbles causes the density of the flocs to decrease to a density below that of the water causing the flocs to float on the water surface.✓ The flocs that float on the surface of the water are removed mechanically or hydraulically.✓ A skimmer is used to remove or skim off the floating flocs.✓ (9)

[36]

TOTAL: 100

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